

Chrysostomos Karakasis

Senior Robotics & Controls Engineer

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Professional Experience

Senior Robotics & Controls Engineer

Manager: Sree Shankar Satheesh Babu

Johnson & Johnson Medtech

July 2024 - PRESENT

- Improved algorithms for the detection and engagement of sterile adapters and tools for robotic surgery
- Developed algorithms for the control of tools for robotic surgery

Robotics and Controls Engineer Intern

Manager: Douglas Spencer Maughan - Mentor: Michael Dermksian

Johnson & Johnson Medtech

June - Aug. 2022

- Designed, implemented, and evaluated a hard-stop handling method to prevent overloading in cable-driven continuum wrists
- Studied the modeling/control of redundant robotic manipulators, covering kinematics and Jacobian-based space transitions
- Contributed code to production C++ robotics code-base and gained experience in unit testing and Git version control

Research Assistant

Advisor: Dr. Panagiotis Artemiadis

University of Delaware

2019 - 2024

- Implemented an admittance controller for ankle-foot prostheses to improve walking stability over compliant terrains
- Proposed an energy-based framework for the dynamic locomotion of bipeds across a wide range of compliant surfaces
- Developed a bio-inspired controller for a biped to regulate expected perturbations of extremely low ground stiffness
- Introduced a novel kinematic data-based algorithm (F-VESPA) for real-time foot-strike detection
- Designed a user-friendly interface for performing gait experiments using the Variable Stiffness Treadmill (VST)

Research Assistant

Advisor: Dr. Dimitrios Soudris & Dr. Evangelos Papadopoulos (MicroLab & CSL-EP)

National Technical University of Athens

2018-2019

- Implemented a highly affordable control architecture for the quadruped robot Laelaps II based on a SoC FPGA

Education

Ph.D. in Mechanical Engineering

UNIVERSITY OF DELAWARE, USA

GPA: 4.0/4.0

AUG. 2019 - JULY 2024

- **Dissertation:** "Robust and Stable Locomotion of Lower-limb Prostheses and Bipedal Robots over Compliant Terrains"
Advisor: Dr Panagiotis Artemiadis
Committee: Ioannis Poulakakis (UD); Thomas Buchanan (UD); Robert Gregg (UMich)

M.Sc. in Mechanical Engineering

UNIVERSITY OF DELAWARE, USA

GPA: 4.0/4.0

DEC. 2023

B.Sc. & M.Sc. in Electrical and Computer Engineering (Top 5%)

NATIONAL TECHNICAL UNIVERSITY OF ATHENS, GREECE (5-YEAR JOINT DEGREE; 300 ECTS)

GPA: 9.02/10

DEC. 2013 - FEB. 2019

- **Thesis:** "Implementation of Quadruped Robot's Motion Control on SoC FPGA"
Supervisors: Prof. Dimitrios Soudris & Prof. Evangelos Papadopoulos

Technical Skills

- **Operating Systems:** Microsoft Windows, Unix (Linux)
- **Programming Languages:** C/C++, Python, Pascal, Assembly (ARM, AVR architecture)
- **Computer Hardware Design:** VHDL, Verilog, Xilinx Vivado/ISE Design Suite, High-Level Synthesis (HLS)
- **Software Tools:** Mathworks MATLAB/Simulink, Visual Studio, Webstorm, GitHub, Bitbucket, Vicon Nexus SDK
Gazebo, Eclipse IDE, Matplotlib
- **Application Software:** T_EX (L^AT_EX, B_IB_TE_X), Microsoft Office, OpenOffice

Academic Scholarships and Awards

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| - IEEE/RAS Member Support Program for IROS 2023 | IEEE Robotics and Automation Society, 2023 |
| - Graduate Student Travel Award for IROS 2023 | University of Delaware, 2023 |
| - George W. Laird Merit Fellowship (Finalist) | University of Delaware, 2020 |
| - Onassis Foundation Scholarship | Alexander S. Onassis Foundation, 2020-2023 |
| - Gerondelis Foundation Scholarship for Academic Excellence | Gerondelis Foundation Inc., 2020 |

Publications

- [J5] **C. Karakasis**, C. Scully, R. Salati, and P. Artemiadis, "Control of powered ankle-foot prostheses on compliant terrain: A quantitative approach to stability enhancement," *Actuators*, vol. 15, no. 2, 2026, ISSN: 2076-0825. DOI: 10.3390/act15020107. [Online]. Available: <https://www.mdpi.com/2076-0825/15/2/107>.
- [J4] C. Angelidou, V. Chambers, B. Hobbs, **C. Karakasis**, and P. Artemiadis, "Kinematics, kinetics, and muscle activations during human locomotion over compliant terrains," *Scientific Data*, vol. 12, no. 1, p. 84, 2025.
- [J3] V. Chambers, B. Hobbs, W. Gaither, Z. Thé, A. Zhou, **C. Karakasis**, and P. Artemiadis, "The Variable Stiffness Treadmill 2: Development and Validation of a Unique Tool to Investigate Locomotion on Compliant Terrains," *Journal of Mechanisms and Robotics*, vol. 17, no. 3, p. 031005, Sep. 2024, ISSN: 1942-4302. DOI: 10.1115/1.4066173. eprint: https://asmedigitalcollection.asme.org/mechanismsrobotics/article-pdf/17/3/031005/7374386/jmr_17_3_031005.pdf. [Online]. Available: <https://doi.org/10.1115/1.4066173>.
- [J2] **C. Karakasis**, I. Poulakakis, and P. Artemiadis, "An Energy-Based Framework for Robust Dynamic Bipedal Walking Over Compliant Terrain," *Journal of Dynamic Systems, Measurement, and Control*, pp. 1–12, Nov. 2023, ISSN: 0022-0434. DOI: 10.1115/1.4064094.
- [J1] **C. Karakasis** and P. Artemiadis, "Real-time kinematic-based detection of foot-strike during walking," *Journal of Biomechanics*, vol. 129, p. 110849, 2021, ISSN: 0021-9290. DOI: <https://doi.org/10.1016/j.jbiomech.2021.110849>.
- [C4] **C. Karakasis**, R. Salati, and P. Artemiadis, "Adjusting the Quasi-Stiffness of an Ankle-Foot Prosthesis Improves Walking Stability during Locomotion over Compliant Terrain," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 2140–2145. DOI: 10.1109/IROS55552.2023.10342344.
- [C3] **C. Karakasis**, I. Poulakakis, and P. Artemiadis, "Robust Dynamic Walking for a 3D Dual-SLIP Model under One-Step Unilateral Stiffness Perturbations: Towards Bipedal Locomotion over Compliant Terrain," in *2022 30th Mediterranean Conference on Control and Automation (MED)*, 2022, pp. 969–975. DOI: 10.1109/MED54222.2022.9837236.
- [C2] **C. Karakasis** and P. Artemiadis, "F-VESPA: A Kinematic-based Algorithm for Real-time Heel-strike Detection During Walking," in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2021, pp. 5098–5103. DOI: 10.1109/IROS51168.2021.9636335.
- [C1] **C. Karakasis**, K. Machairas, C. Marantos, I. S. Paraskevas, E. Papadopoulos, and D. Soudris, "Exploiting the SoC FPGA Capabilities in the Control Architecture of a Quadruped Robot," in *2020 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*, IEEE, 2020, pp. 501–507. DOI: 10.1109/AIM43001.2020.9159012.

Professional Service

Reviewer for journal and conference submissions:

- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2022 - present
- Frontiers in Robotics and AI, 2021 - present
- International Journal of Robotics Research (IJRR), 2021 - present
- IEEE Transactions on Neural Systems & Rehabilitation Engineering (TNSRE), 2020 - present
- ASME Journal of Mechanisms and Robotics (JMR), 2019 - present
- IEEE International Conference on Robotics and Automation (ICRA), 2019 - present
- Robotica, 2021 - present
- Wearable Technologies, 2020 - present
- IEEE Robotics and Automation Letters (RA-L), 2023 - present
- IEEE International Conference on Rehabilitation Robotics (ICORR), 2023 - present
- American Control Conference (ACC), 2023 - present
- PLOS ONE, 2024 - present